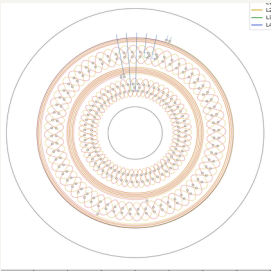


INDUCTIVE ENCODER – SIGNAL BOARD (rev A)

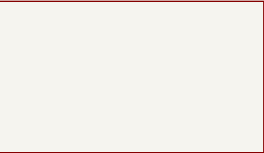
Contactless absolute rotary encoder after TI TIDA-010961 (TIDUFF8).
Dual-track Nonius coils (16 / 15 periods) on a 4-layer Ø76 mm stator, Ø56 mm
copper target disc, 0.5 mm air gap. Two LDC5072-Q1 AFEs excite the coils and
demodulate sin/cos; MSPM0G3507 samples both ADCs simultaneously and computes
the absolute angle (Nonius sector + fine angle), sent over UART.

KEY SPECS
Input: 12-24 V (4-65 V buck), ~0.5 W
Coil area: Ø58 mm, shaft hole Ø16 mm
Tracks: 16 periods (outer), 15 periods (inner)
Excitation: ~4.7 MHz (outer), ~3.85 MHz (inner)
MCU: MSPM0G3507, 2x 12-bit 4 Msps ADC
Interface: UART 2.5 Mbaud, SYNC input, SWD
Target accuracy (TI ref.): < 0.04°, 16+ ENOB



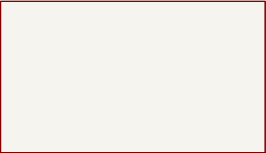
Stator coils (generated)

Power



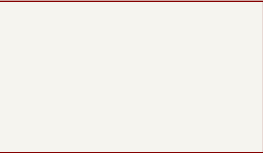
File: power.kicad_sch

Interface



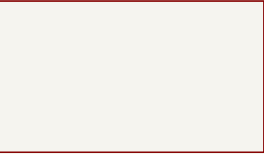
File: interface.kicad_sch

MCU



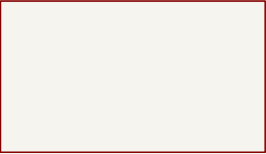
File: mcu.kicad_sch

AFE_Outer



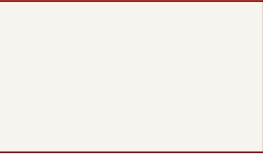
File: afe_outer.kicad_sch

AFE_Inner



File: afe_inner.kicad_sch

Coil_Connectors



File: coil_connectors.kicad_sch

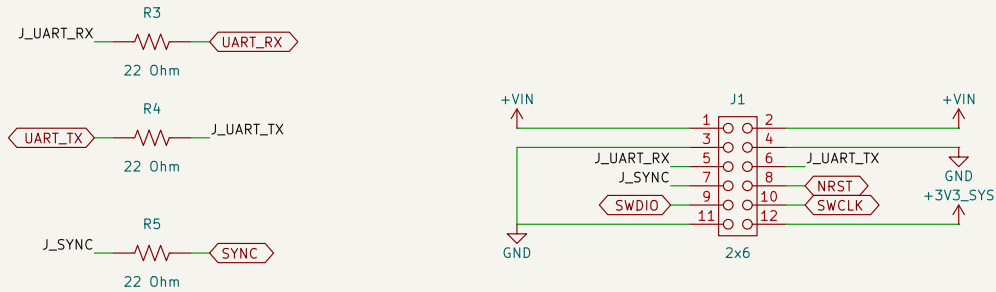
REVISION HISTORY
A 2026-10-09 first issue

Title: Inductive Encoder – Signal Board			
Project: Inductive Encoder			
Number: TBD	Rev: A		
Sheet: /	File: EncoderSignal.kicad_sch	Designer : Ekrem Turan Firat	
Size: A4	Date: 2026-10-09		

TEST POINTS

Designer : Ekrem Turan Fırat

ENCODER CONNECTOR J1 (2x6, 2.54 mm)

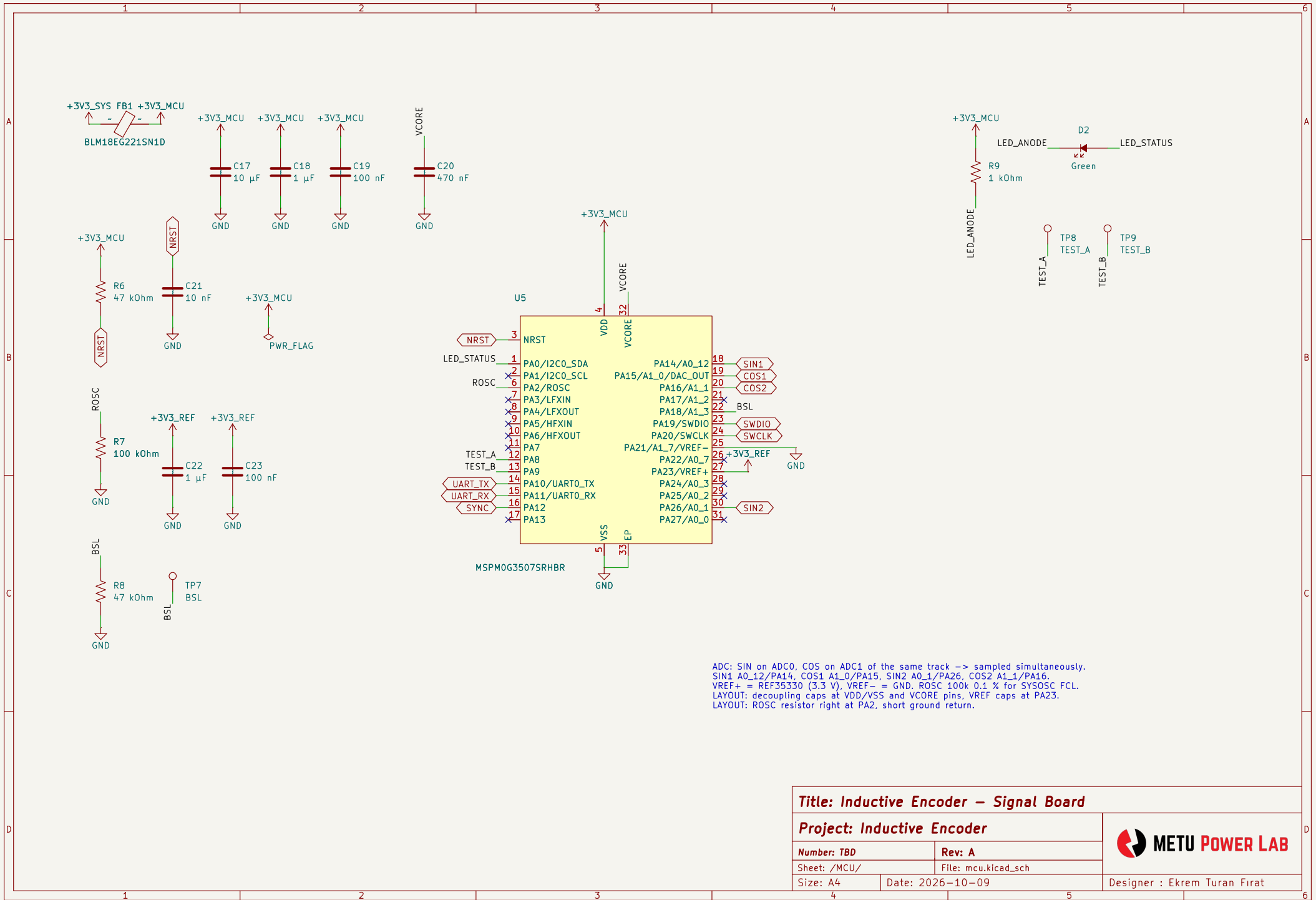


Pinout
1,2 +VIN 12-24 V 3,4,11 GND
5 UART_RX (host -> encoder) 6 UART_TX (encoder -> host)
7 SYNC (sample trigger in) 8 NRST
9 SWDIO 10 SWCLK 12 +3V3 (debugger Vtref. out)
UART: 2.5 Mbaud, 8N1 (TIDA-010961 protocol). The MSPM0 ROM bootloader
also uses UART0 (PA10/PA11): hold BSL (PA18) high at reset to flash over J1.

LAYOUT: J1 on the bottom side (away from the target), at the board edge.

Title: Inductive Encoder – Signal Board		
Project: Inductive Encoder		
Number: TBD	Rev: A	
Sheet: /Interface/	File: interface.kicad_sch	
Size: A4	Date: 2026-10-09	Designer : Ekrem Turan Firat



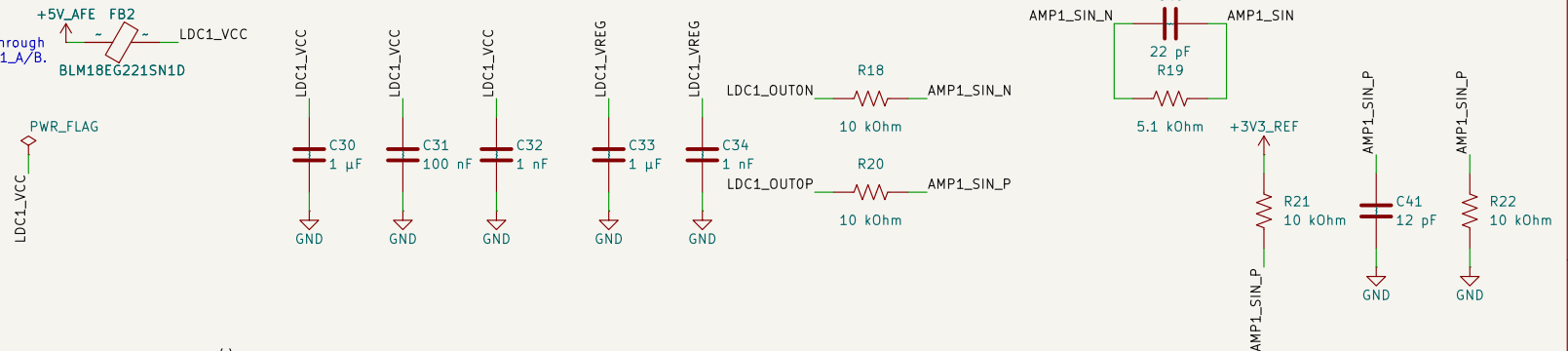


Title: Inductive Encoder – Signal Board				
Project: Inductive Encoder				
Number: TBD		Rev: A		
Sheet: /MCU/		File: mcu.kicad_sch		
Size: A4	Date: 2026-10-09			
4		5		Designer : Ekrem Turan Firat

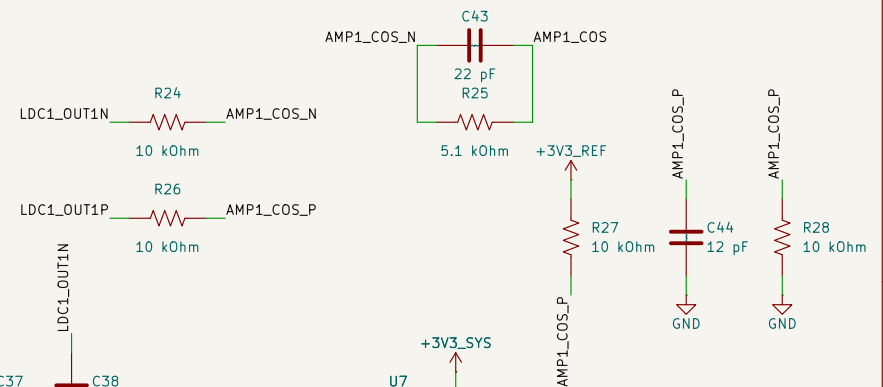
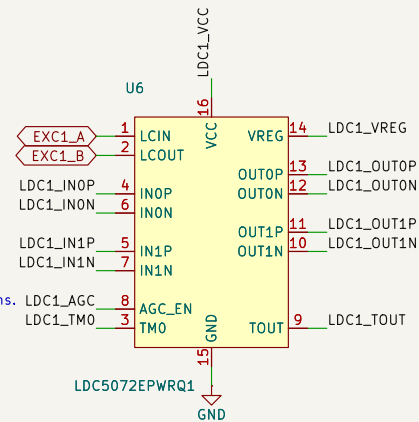
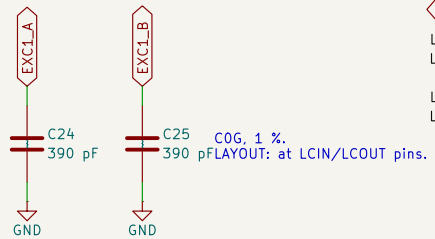
COILS – track 1: 16 periods

The coils are on the stator board. Their terminals arrive through J2/J3 (sheet Coi_Connectors): SIN1_P/N, COS1_P/N, EXC1_A/B.

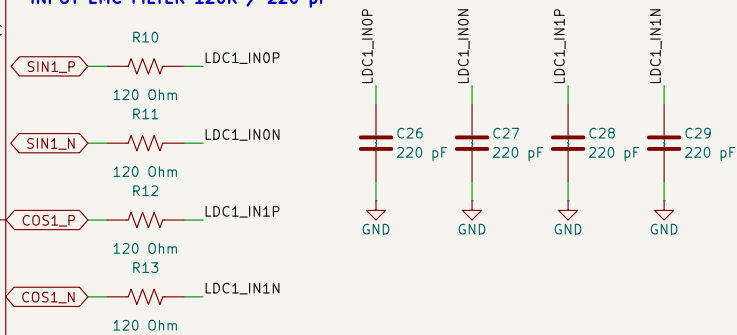
DIFF -> SINGLE-ENDED G = 5.1k/10k, bias 1.65 V



LC TANK $f = \sim 4.7 \text{ MHz}$ ($5.8 \mu\text{H}$)

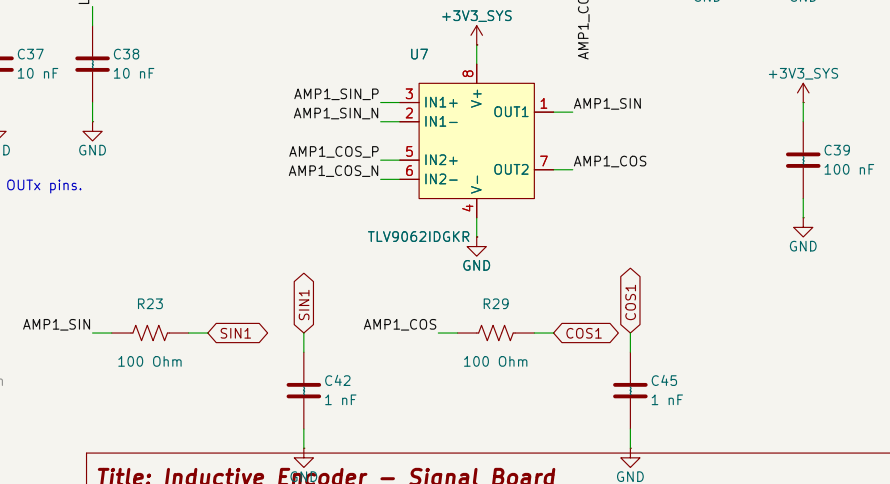


INPUT EMC FILTER 120R / 220 pF

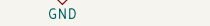


AGC mode: R(AGC_EN->GND) = 1.5k, pull-up DNP.
Fixed gain: fit pull-up, 0.1 % parts (datasheet 9.2.2.2.3).

COUT 10 nF. LAYOUT: at the OUTx pins.

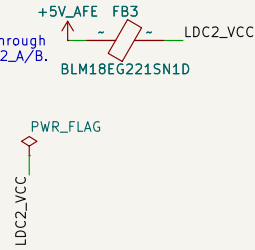


Out = $0.51 \times (\text{OUTxP} - \text{OUTxN}) + 1.65 \text{ V}$. 100R/1 nF anti-alias (1.6 MHz)
LAYOUT: RC right at the MSPM0 ADC pins. Keep SIN/COS routing symmetric.

Title: Inductive Encoder – Signal Board		
Project: Inductive Encoder		
Number: TBD	Rev: A	
Sheet: /AFE_Outer/	File: afe_outer.kicad_sch	
Size: A4	Date: 2026-10-09	Designer : Ekrem Turan Firat

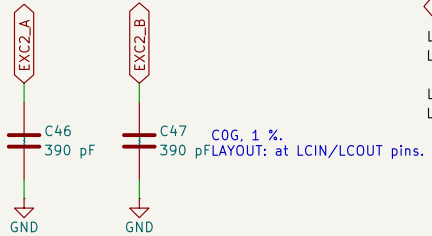
COILS – track 2: 15 periods

The coils are on the stator board. Their terminals arrive through J2/J3 (sheet Coil_Connectors): SIN2_P/N, COS2_P/N, EXC2_A/B.

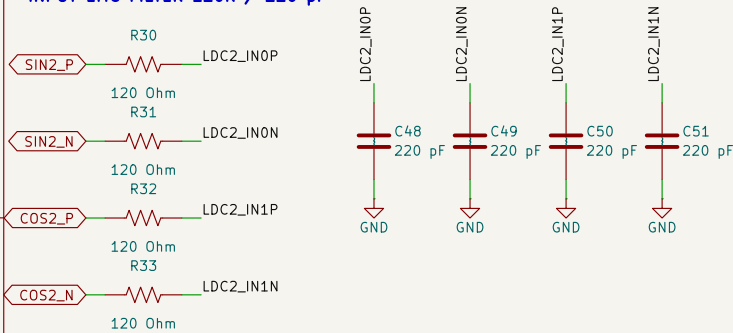


DIFF -> SINGLE-ENDED G = 5.1k/10k, bias 1.65 V

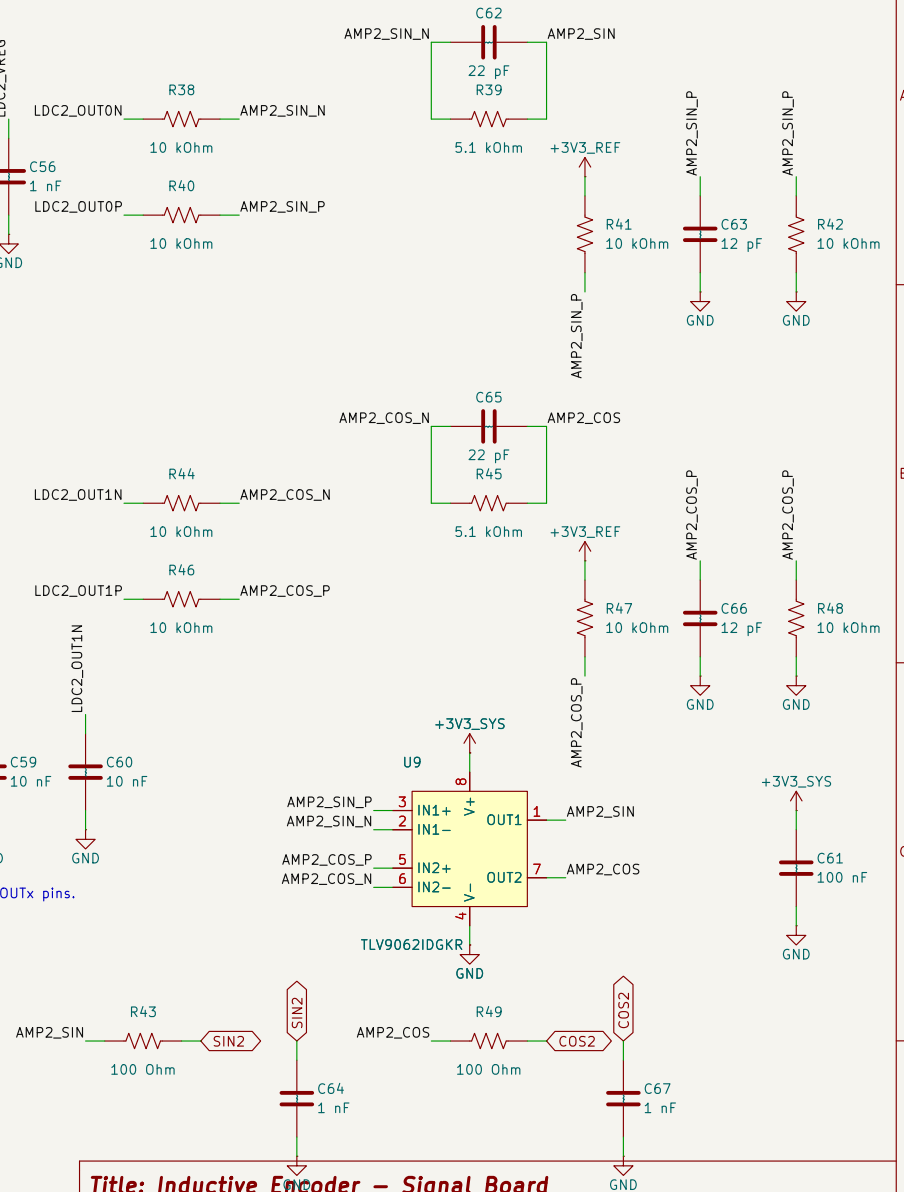
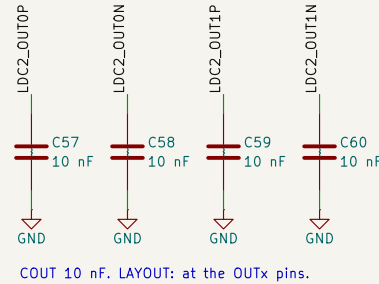
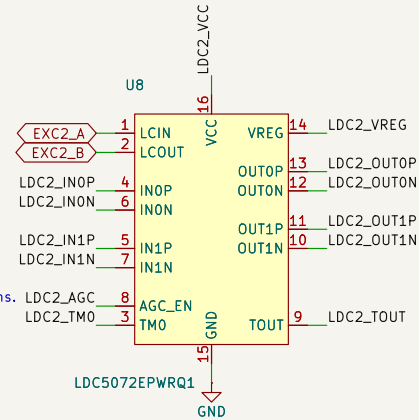
LC TANK $f = -3.85$ MHz (8.8 μ H)



INPUT EMC FILTER 120R / 220 pF



AGC mode: R(AGC_EN->GND) = 1.5k, pull-up DNP.
Fixed gain: fit pull-up, 0.1 % parts (datasheet 9.2.2.2.3).



Title: Inductive Encoder – Signal Board

Project: Inductive Encoder

Number: TBD

Rev: A

Sheet: /AFE_Inner/

File: afe_inner.kicad_sch

Size: A4

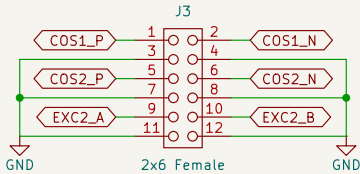
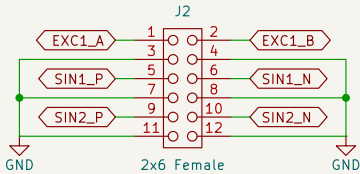
Date: 2026-10-09



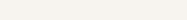
Designer : Ekrem Turan Firat

Out = 0.51 x (OUTxP - OUTxN) + 1.65 V. 100R/1 nF anti-alias (1.6 MHz)
LAYOUT: RC right at the MSPM0 ADC pins. Keep SIN/COS routing symmetric.

COIL CONNECTORS J2 / J3 (2x6 2.54 mm sockets, mate with the stator headers)



J2: EXC1 (outer excitation), SIN1, SIN2. J3: COS1, COS2, EXC2.
Every second pin pair is GND to separate the coil pairs.
LAYOUT: J2/J3 on the TOP side, at the same X/Y as the stator headers
(stacked on M3 standoffs). Keep each P/N pair together to the AFE.

Title: Inductive Encoder – Signal Board			
Project: Inductive Encoder			
Number: TBD		Rev: A	
Sheet: /Coil_Connectors/		File: coil_connectors.kicad_sch	
Size: A4	Date: 2026-10-09		
Designer : Ekrem Turan Firat			